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The Influence of Cognitive Styles on Visual Search Efficiency in Eastern and Western Interface Designs: An Eye-Tracking Study

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INTRODUCTION



- **Theoretical Foundation:** Nisbett's cognitive framework (2001) contrasts Western analytic style (focal focus) with East Asian holistic style (contextual relationship focus).
- **The Core Clash of Models:**
 - Processing Advantage (Wang et al., 2012): Holistic users process information-dense designs faster.
 - Global Intake Cost (Baughan et al., 2021): Holistic users scan pages globally first, causing search delays.
- **Methodological Gap:** Prior studies compared cultures as country groups, confounding cognitive variables ---> My study addresses this gap by measuring cognitive style at the individual level within a single, homogeneous Polish sample

RESEARCH AIMS & HYPOTHESES



- **Primary Objective:** The present study aimed to investigate whether cognitive style, as measured by the AHS-24, moderates visual search efficiency on Eastern and Western web interfaces differing in visual complexity.
- **H1:** Response times will be significantly shorter for Western interfaces than for Eastern interfaces across all participants, reflecting the higher visual complexity of Eastern interfaces.

RESEARCH AIMS & HYPOTHESES



- **H2:** Cognitive style will moderate response times as a function of interface type: participants with higher holistic scores on the AHS-24 will demonstrate shorter response times on Eastern interfaces than participants with higher analytic scores, whereas the reverse pattern will be observed on Western interfaces.
- **H3:** Cognitive style will moderate fixation rate (fixations per second) as a function of interface type: more holistic participants will produce a lower fixation rate on Eastern interfaces compared to more analytic participants, whereas more analytic participants will produce a lower fixation rate on Western interfaces compared to more holistic participants.

METHODOLOGY

- Participants:
 - n=50
 - age: 18 to 31 years old (M = 21.58)
 - Poles (mostly students), with no knowledge of Japanese
- Apparatus: Gazepoint GP3 eye-tracker
- Within-subject design
- Stimuli: 48 trials in total: 8 western interfaces, 8 eastern interfaces x 3 (spades, diamonds, clubs)
- Software: OGAMA 5.1
- Demographic questionnaire and AHS-24 in Google Forms
- Analysis: Jamovi and R

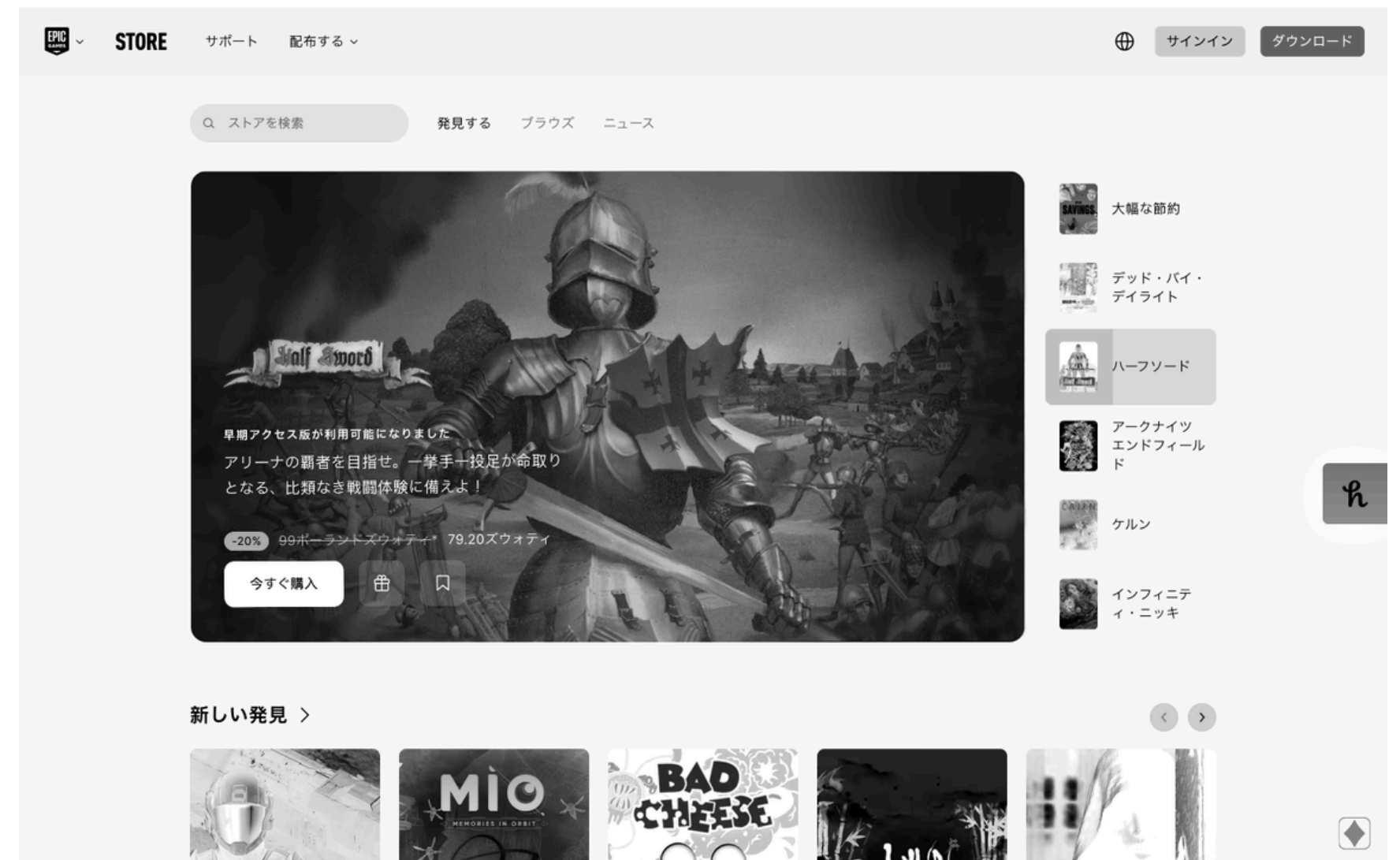
STIMULI



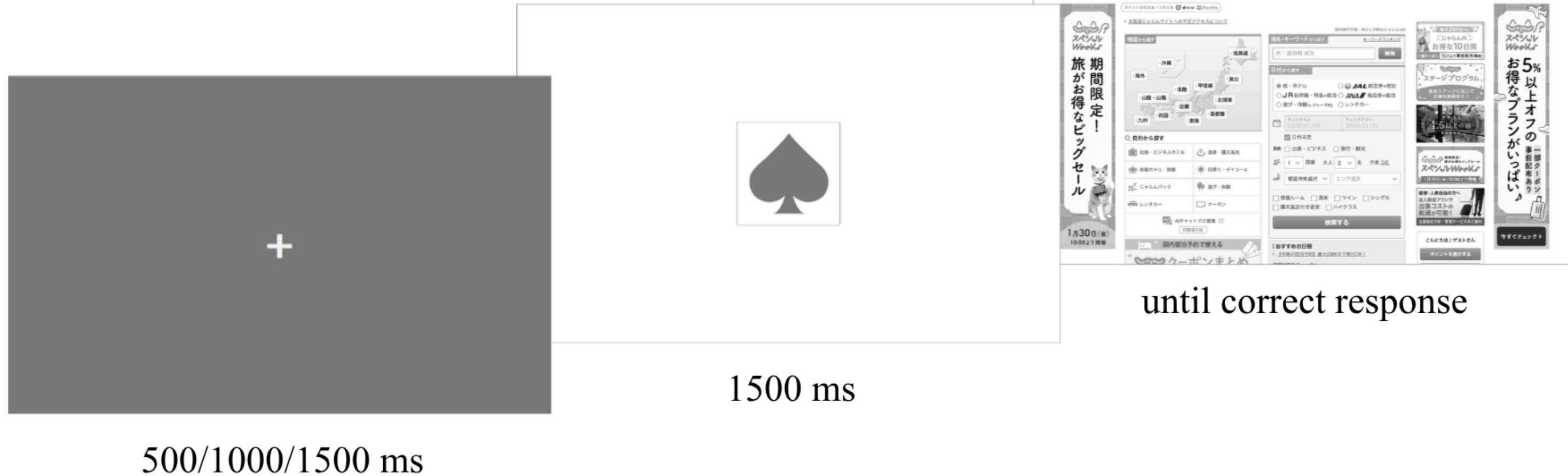
Category	Eastern Interface	Western Interface
Car rental	2525r.com	govirtuo.com
Travel and hospitality	jalan.net	splendia.com
Gaming and entertainment	dmm.com	store.epicgames.com
Transportation	jorudan.co.jp	citymapper.com
Online marketplace	rakuten.co.jp	shop.app
Higher education	titech.ac.jp	utexas.edu
Auction platforms	auctions.yahoo.co.jp	artsy.net
Secondhand goods	oikura.jp	rubylane.com

STIMULI

- Perceived visual complexity model described by Reinecke et al. (2013)
- **Eastern** (complexity score M = 611.00)
- **Western** (complexity score M = 147.96)



PROCEDURE



- 2 blocks corresponding to interface type

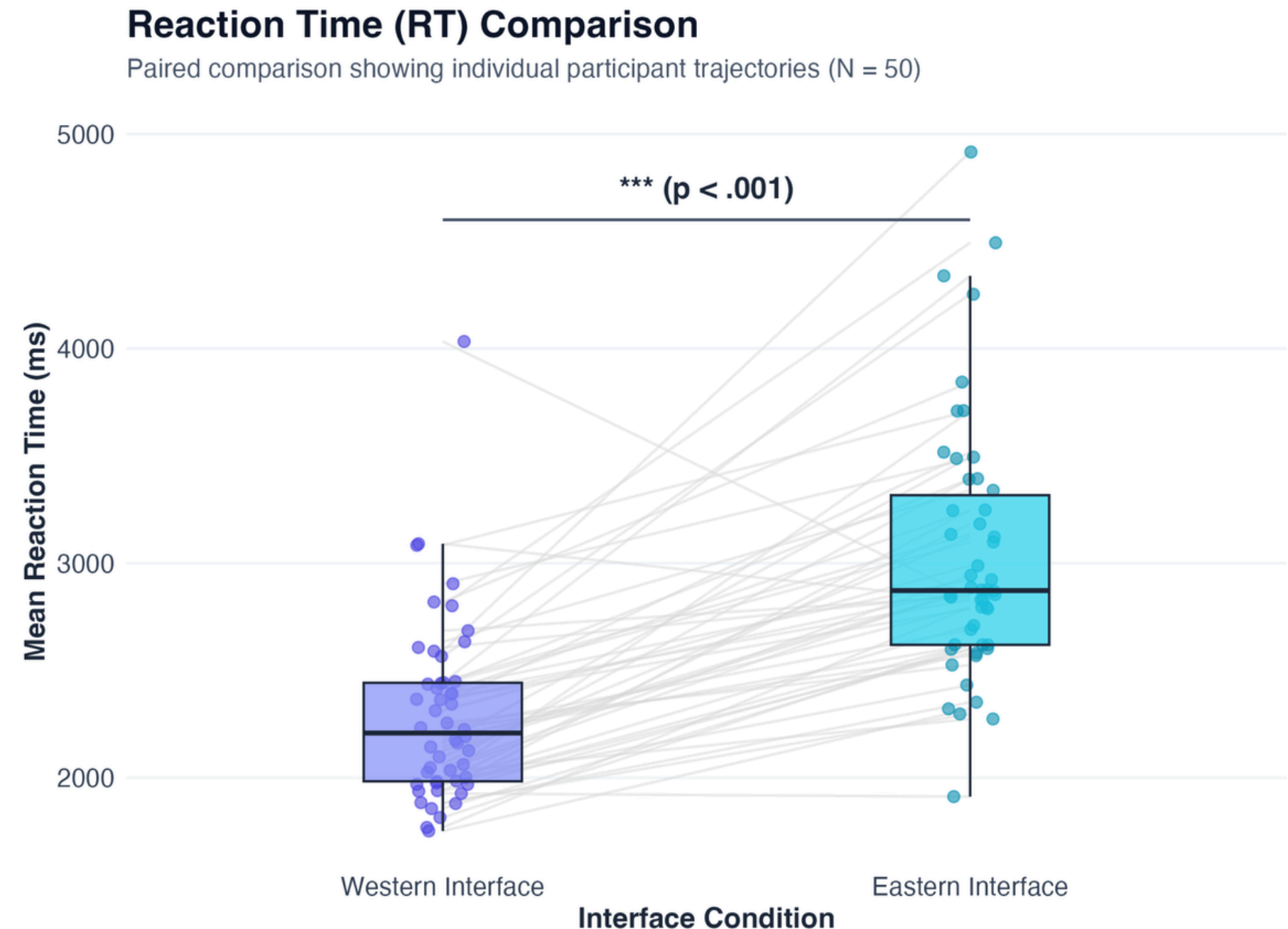
ANALYSIS-HOLISM SCALE (AHS-24)

- **Measurement Tool:** 24-item questionnaire (Choi, Koo, & Choi, 2007) assessing analytic vs. holistic style.
- **Four Cognitive Dimensions:**
 - **Causality:** Belief in complex, interconnected causes (holistic) vs. linear causes (analytic).
 - **Locus of Attention:** Global visual field focus (holistic) vs. focal object focus (analytic).
 - **Perception of Change:** Viewing reality as cyclical/changing (holistic) vs. static (analytic).
 - **Attitude toward Contradictions:** Dialectical reconciliation (holistic) vs. polarized logic (analytic).

**M = 122.40, SD = 11.54,
range: 92.00–140.00**

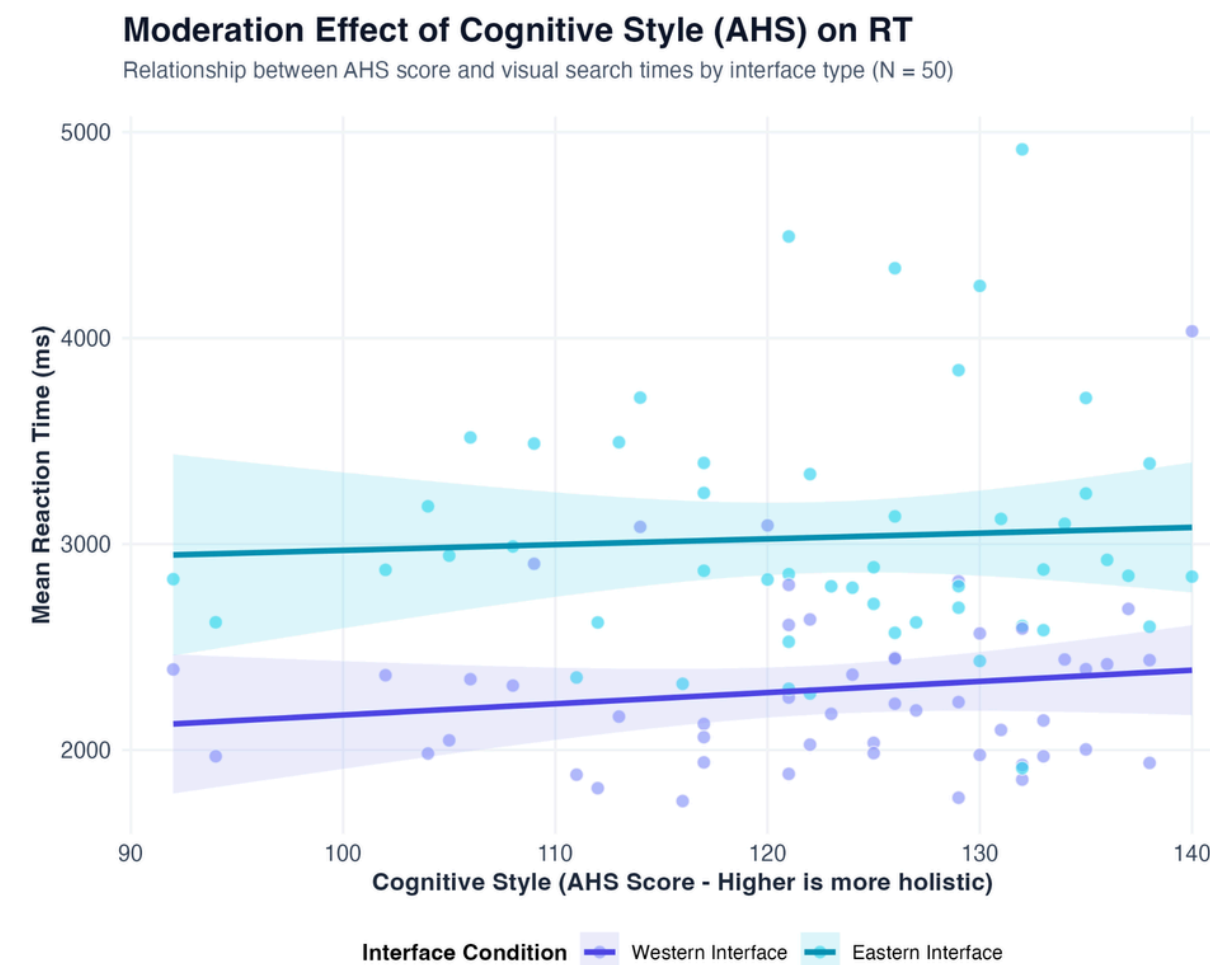
RESULTS: H1 SUPPORTED

- **Reaction Time:** Shorter search times on Western than Eastern interfaces ($t(49) = -10.95$, $p < .001$, Cohen's $d = 1.55$).
- **Western Mean RT:** 2,292.15 ms (SD = 397.69)
- **Eastern Mean RT:** 3,031.84 ms (SD = 585.83)
- **Visual Complexity Cost:** Eastern layouts impose a significant, universal processing cost (approx. 740 ms delay) across all participants.

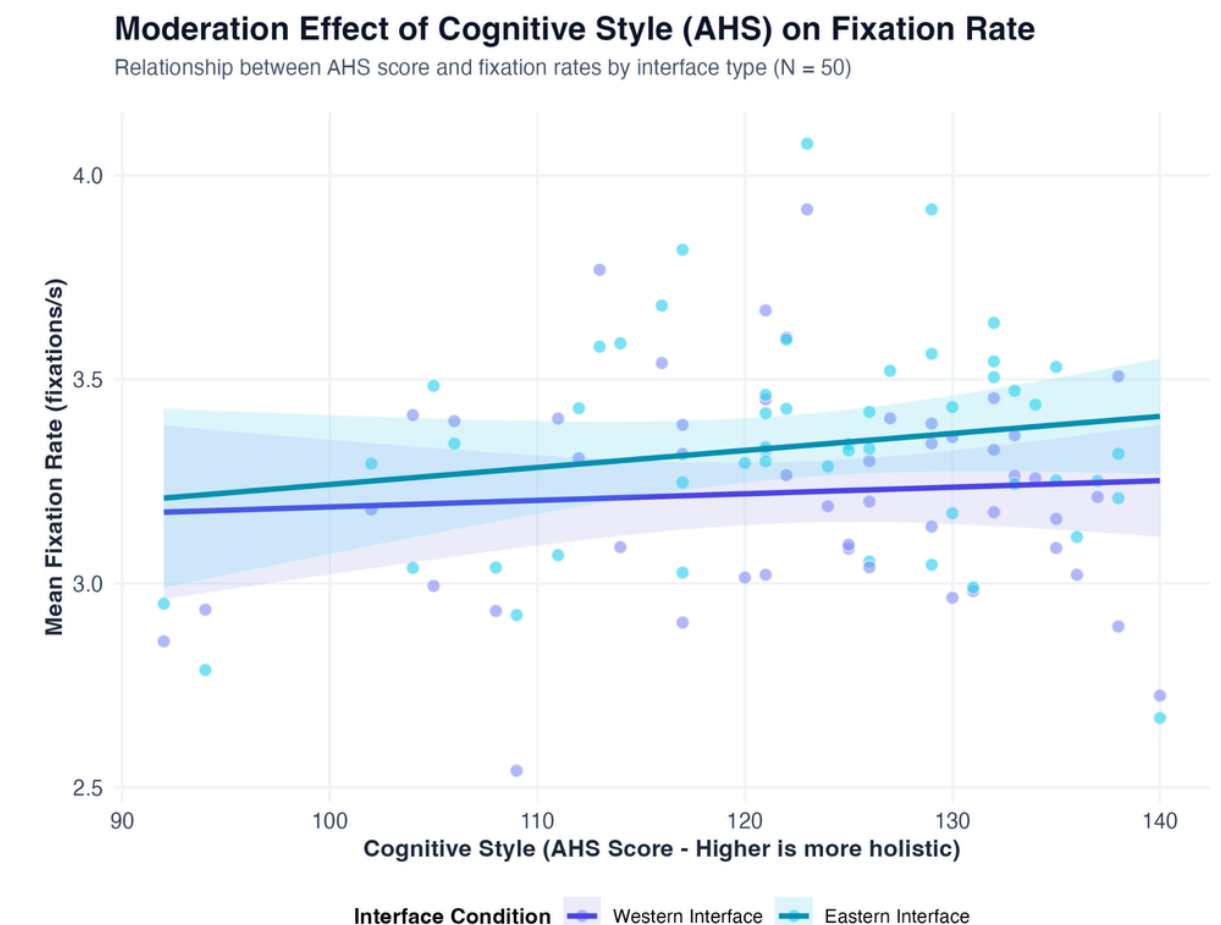


RESULTS: H2 & H3

- **Reaction Time Moderation (H2):** Composite AHS-24 score did not moderate RT. The Interaction term was non-significant: $F(1, 48) = 0.41$, $p = .526$, $\eta^2p = .008$.



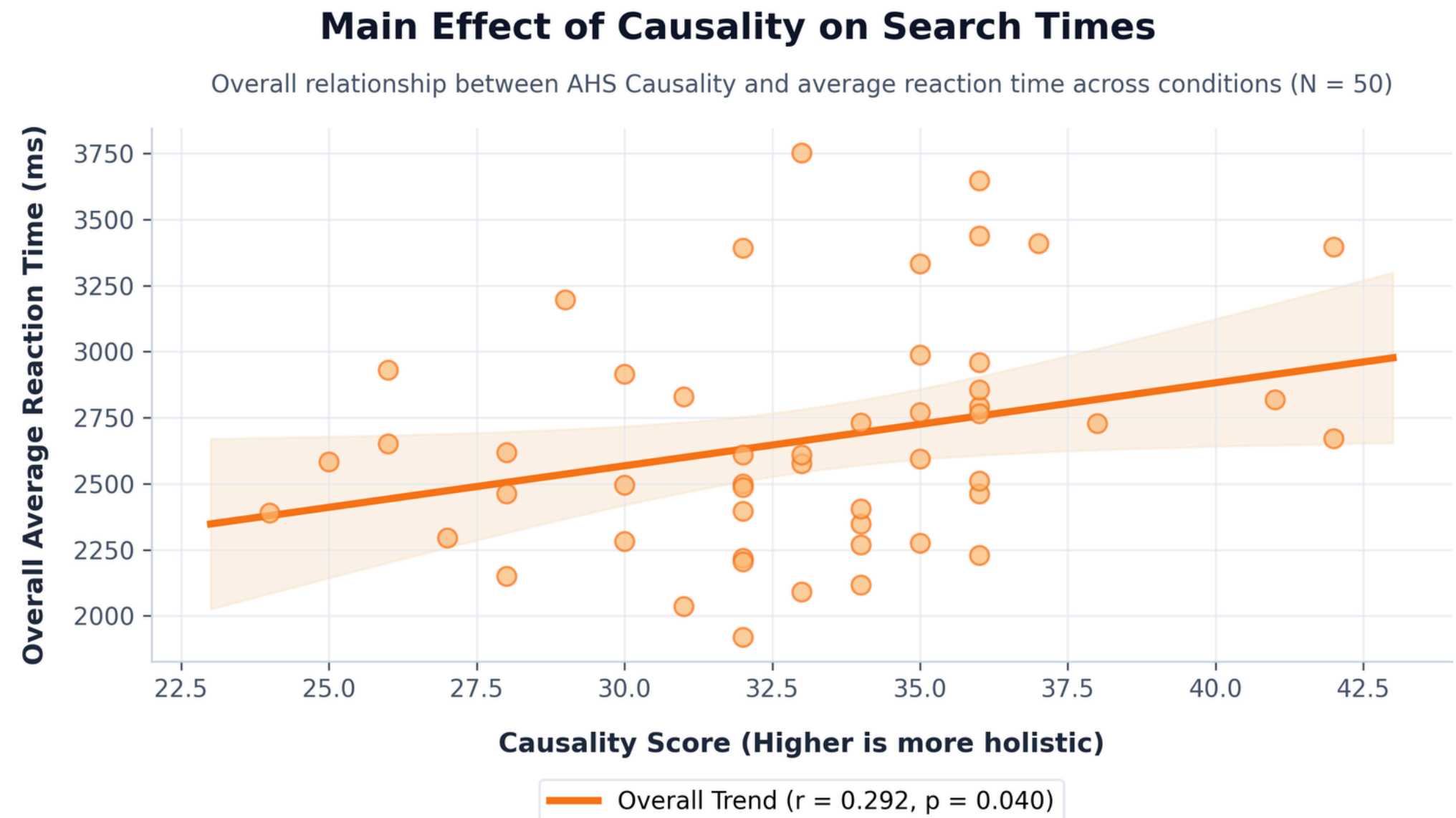
- **Fixation Rate Moderation (H3):** Composite AHS-24 score did not moderate fixation rates. Interaction term: $F(1, 48) = 0.90$, $p = .346$, $\eta^2p = .019$.



- individual variations in overall cognitive style do not moderate visual search efficiency in speeded tasks

EXPLORATORY: CAUSALITY SUBSCALE

- **Causality Main Effect:**
Significant overall effect on Reaction Time ($F(1, 48) = 4.55$, $p = .038$, $\eta^2_p = .087$). Holistic causality (belief in interconnectedness) is associated with slower visual search times overall.
- **Western Condition:**
statistically significant positive correlation ($r = .290$, $p = .041$).
- **Eastern Condition:**
non-significant positive correlation ($r = .219$, $p = .127$).



----> cognitive mismatch

Discussion: Why Null Moderation?

- **Sample Homogeneity & Range Restriction:** AHS-24 score distribution was restricted (range: 92–140 vs. theoretical: 24–168). A culturally homogeneous Polish sample limits cognitive style variance compared to cross-national groups.
- **Perceptual Load Ceiling Effect:** The extreme visual density of Eastern layouts ($M = 611$) demanded substantial attention resources from all users, leaving little room for cognitive style to explain performance variance.
- **Digital Native Standardization:** Young-adult students employ highly automated, task-focused visual navigation habits (functional search grids) that override subtle cognitive style differences during speeded search.

Discussion: Theoretical and Practical Contributions

- **Style is Not a Monolith:** Aggregating dimensions into a composite AHS-24 score masked results. Individual subscale analysis (specifically Causality) was essential to reveal cognitive patterns.
- **Empirical Support for Global Intake Cost:** The significant Causality main effect provides individual-level proof of Baughan's theory: holistic reasoning search delay is caused by relationally scanning the scene as an interconnected whole.
- **Universal Value of Simplicity:** Visual simplicity (Western style) enhances search efficiency for all users, regardless of cognitive style. Clean grids, clear hierarchies, and white space should be prioritized.

Limitations & Future Directions

- **Limitations:**

- Homogeneous young-adult sample (mostly Polish students).
- Entry-level 60 Hz eye-tracker (Gazepoint GP3) without head stabilization (chin rest) introduces minor spatial/temporal noise.
- Static screenshots instead of fully interactive, scrollable web environments.
- Focal search task ignores other web behaviors (like scanning, aesthetic reading, free-browsing).

- **Future Directions:**

- Direct cross-cultural eye-tracking comparison (Polish vs. East Asian participants) to increase AHS-24 variance.
- Utilizing interactive layouts and scrollable mockups to capture real-time navigation.

THANK YOU!